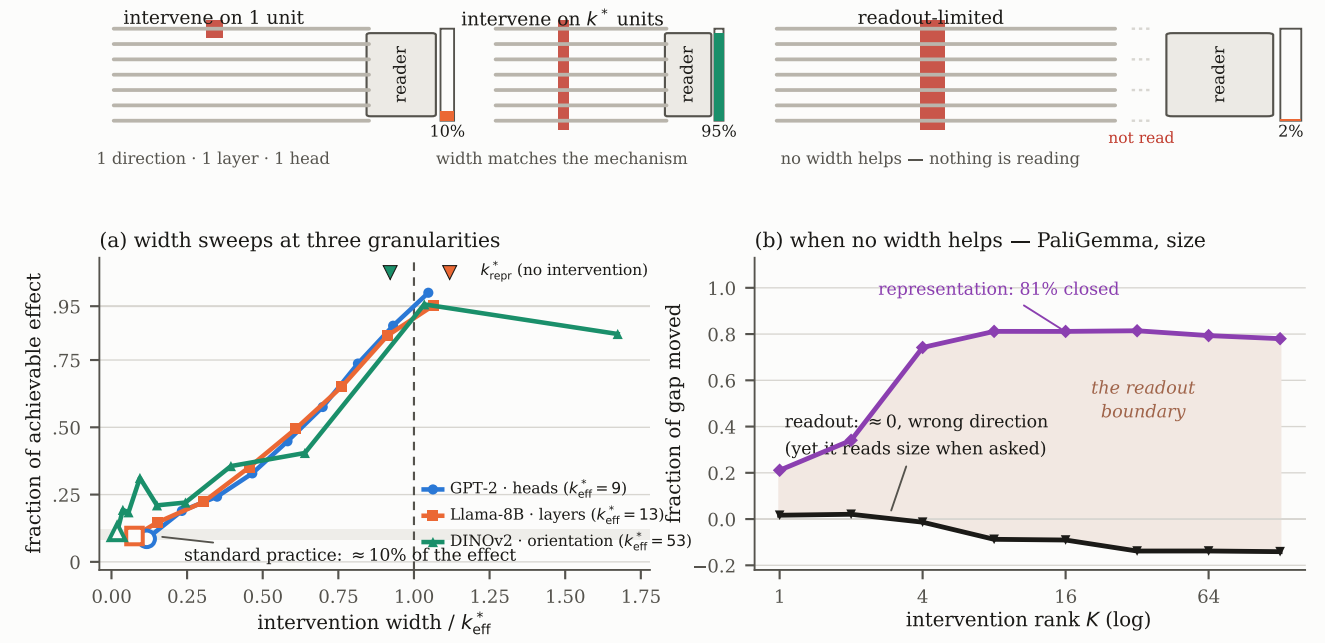
intervene on 1 unit
reader
10%
1 direction · 1 layer · 1 head
intervene on k^* units
reader
95%
width matches the mechanism
readout-limited
reader
not read
2%
no width helps — nothing is reading
(a) width sweeps at three granularities
fraction of achievable effect
k^*_{repr} (no intervention)
GPT-2 · heads ($k^*_{eff} = 9$)
Llama-8B · layers ($k^*_{eff} = 13$)
DINOv2 · orientation ($k^*_{eff} = 53$)
standard practice: ≈ 10% of the effect
intervention width / k^*_{eff}
(b) when no width helps — PaliGemma, size
representation: 81% closed
the readout boundary
readout: ≈ 0, wrong direction
(yet it reads size when asked)
fraction of gap moved
intervention rank K (log)